BIAS ANALYSIS REPORT

1. Sensitive Feature Identification

Out of the 77 medical and demographic parameters utilized as Machine Learning inputs, several sensitive and structural features possess the potential to introduce algorithmic bias if not properly addressed. These features include:

- **Target Class (Class Label):** The sepsis_label column indicating the patient's condition.
- **Demographics & Background:** Columns such as gender, ethnicity, and age.
- **Socioeconomic Status:** The insurance column (Health Insurance Type).
- **Synthetic Generation Bias:** The artificial mathematical rules from the dataset that separate clinical columns (e.g., pao2_fio2_ratio) perfectly without realistic overlapping.

2. Descriptive Statistics

Based on direct extraction from the population dataset of 5,000 patients, the distribution of sensitive features exhibits the following imbalances:

- **Class Imbalance (Sepsis vs. Safe):** Patients with a safe status dominate the dataset at **85.0%** (4,250 patients), while positive Sepsis cases constitute the minority class at only **15.0%** (750 patients).
- **Gender Distribution:** Male patients (M) account for **56.4%** and Female (F) for **43.6%**.
- **Racial/Ethnic Distribution:** The majority of the data is represented by the Caucasian/White demographic (**59.5%**), while other groups constitute minorities: Black (15.6%), Hispanic (11.9%), Asian (7.9%), and Other (5.1%).
- **Patient Age:** The average age of the population is **64 years**, indicating that the data is predominantly skewed toward the elderly (geriatric) patient group.

3. Impact Analysis

If these imbalances and sensitive features are left unmitigated during AI training, the consequences could be detrimental in real-world clinical applications:

- **Impact of Class Imbalance:** The AI will fall into *Majority Class Bias*. The model will consistently predict "Safe Patient" merely to achieve an accuracy above 80%. Consequently, *False Negatives* will spike (the AI declares safe when the patient actually has Sepsis), leading to potential loss of life due to a delay of care.
- **Impact of Demographics & Socioeconomics:** If the AI utilizes ethnicity or insurance as determining factors, patients from minority backgrounds or lower-tier insurance plans (e.g., Medicaid) risk receiving unequal or marginalized ICU care recommendations.
- **Impact of Synthetic Bias:** The perfectly drawn boundaries for oxygen ratios by the artificial data cause the AI to yield absolute certainty levels (0% or 100%). In a highly

ambiguous real-world clinical setting, this overconfidence could cause the AI to miss the gray-area transitions into septic shock.

4. Mitigation Actions

To neutralize these biases and establish a fair Clinical Decision Support System (Algorithmic Fairness), this research implements the following mitigation strategies:

1. **Algorithmic Class Bias Mitigation:** Utilizing the XGBoost algorithm with the **scale_pos_weight** parameter. This approach assigns a significantly heavier penalty to the AI if it misclassifies the minority class (Sepsis), forcing the model to prioritize infection detection sensitivity rather than merely guessing the majority safe class.
2. **Objective Decision Isolation (Demographic Fairness):** Ensuring that the *Feature Importance* (root decision determinants) in XGBoost relies purely on absolute physiological indicators such as *Lactic Acid* (lactate_mmol) and *Oxygen Ratio* (pao2_fio2_ratio). Thus, metrics like age, ethnicity, and insurance status do not discriminatorily influence the medical verdict.
3. **Implementation of Hybrid Reasoning Architecture:** Establishing standard medical rules (qSOFA Score) as a Safety Net. If the AI experiences a blind spot or is overly confident that a patient is healthy due to synthetic data bias, the qSOFA logic on the backend will trigger an automatic override if the patient physically exhibits signs of cardiovascular collapse.
4. **System Limitations (Final Evaluation Disclaimer):** It is noted as a research limitation that prior to deploying the model in an actual ICU setting, it must be retrained using a genuine Electronic Medical Record (EMR) database from a local hospital to dismantle the synthetic boundary bias and adapt to the local patient distribution.